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Chaitin's number Omega

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Problem: How can a number be truly "random"?

In 1974, Gregory Chaitin succeeded in defining a random number Ω . This sounds like a contradiction. How can we define something that remains random?

Chaitin's number Omega

CREDITS

**UNDERSTANDING ARTIFICIAL INTELLIGENCE
THROUGH ALGORITHMIC INFORMATION THEORY**
"CHAITIN'S NUMBER Ω "

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that
programs will halt.
And yet,
this knowledge remains
forever
inaccessible to us.
Let's leave the conclusion to
Chaitin:
"[omega] is violently,
maximally uncomputable,
but it almost looks
computable.
It's just across the border
between
what we can deal with and
things

Omega

0/1 point (ungraded)

Why should we consider Ω as random?

- ☐ Because Omega is a probability.
- ☒ Because no digit of Omega is computable. ✓
- ☐ Because although Turing machines are deterministic, their halting is not.
- ☒ Because the computation of some of Omega's digits may loop indefinitely.

✗

Answer

Incorrect:

No digit of Omega is computable, so Omega, though being perfectly defined mathematically, should be considered random.

Note that Turing machines are deterministic, and so is their halting when they are fed with a given program. The only probabilistic aspect in this story is the fact that the programs' bits are supposed to be drawn through a random procedure.

Submit

You have used 1 of 1 attempt

i Answers are displayed within the problem

Question: Does it make sense to "randomly pick any particular real number between 0 and 1"?

Answer: No! If ever you can "pick" it, then it must be simple. But virtually all real numbers are complex.

In mathematics, one is often asked to choose "any" real number, for instance in $[0,1]$. Real numbers between 0 and 1 are in correspondence with binary sequences (finite or infinite) (just consider the binary development of numbers 0.1011000010... to see that). We can therefore talk about the complexity of real numbers.

Suppose you can "pick" a real number r in $[0,1]$. This means that you have a computable procedure p that outputs r . If so, p can be used to define r ; therefore $K(r) < |p|$, where $|p|$ is the length in bits of p . As a result, r (considered as a binary sequence) is compressible down to a finite quantity and is therefore non-random.

We know, however, that most finite binary sequences are incompressible. This result can easily be extended to infinite sequences. All of them are incompressible, but a zero-length subset of them. In other words, the subset of real numbers that are compressible form a zero-length set. This means that you can't pick *any* real number, just the simple ones, which means... virtually nothing.

Definable numbers

0/1 point (ungraded)

Now consider all numbers that can be defined in mathematics (*i.e.* in a theory such as set theory). It includes rational numbers, $\sqrt{42}$, π ... What can we say about the set of definable numbers?

- ☐ Mathematical formulas are like programs, so definable numbers are computable real numbers.
- ☐ There are only countably many mathematical formulas, so most reals cannot be captured by formulas.
✓
- ☒ This set includes all real numbers, even an incompressible number such as Omega.

✗
Answer
Incorrect:
Mathematical formulas act like constraints. When used to define real numbers, the constraint is supposed to have only one solution. So they are not like programs. This is why Chaitin's number Omega is not computable, but is definable. However, since there are only countably many formulas, virtually all reals remain undefinable.

You have used 1 of 1 attempt

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